

Ethics Plan for Healthcare AI

Principles:

Healthcare AI systems should be designed and used according to key ethical principles, including fairness, privacy, transparency, accountability, and safety. Fairness ensures that all patients receive equal consideration regardless of age, gender, ethnicity, or background. Privacy protects sensitive medical information from unauthorized access. Transparency allows healthcare professionals to understand how AI systems make decisions, while accountability ensures responsibility for outcomes generated by AI tools.

Bias in Healthcare:

Bias can occur when AI models are trained on incomplete, inaccurate, or unbalanced healthcare datasets. Such bias may result in unequal treatment recommendations or inaccurate predictions for certain patient groups. To reduce bias, healthcare organizations should regularly evaluate model performance, use diverse and representative datasets, and apply fairness testing throughout the development process.

Case Example:

Consider a patient risk prediction model trained primarily on data from one population group. The model may predict health risks accurately for that group but less accurately for others. This can lead to unfair healthcare decisions. Continuous monitoring, validation, and model updates can help identify and correct such issues.

Regulations:

Healthcare organizations must comply with regulations such as HIPAA (Health

Insurance Portability and Accountability Act), which protects patient privacy and ensures secure storage, sharing, and management of medical information.

Healthcare Application:

Ethical AI can support fair patient prediction, disease risk assessment, and clinical decision-making. By following ethical guidelines, healthcare providers can improve patient outcomes, maintain trust, and ensure responsible use of healthcare technology.